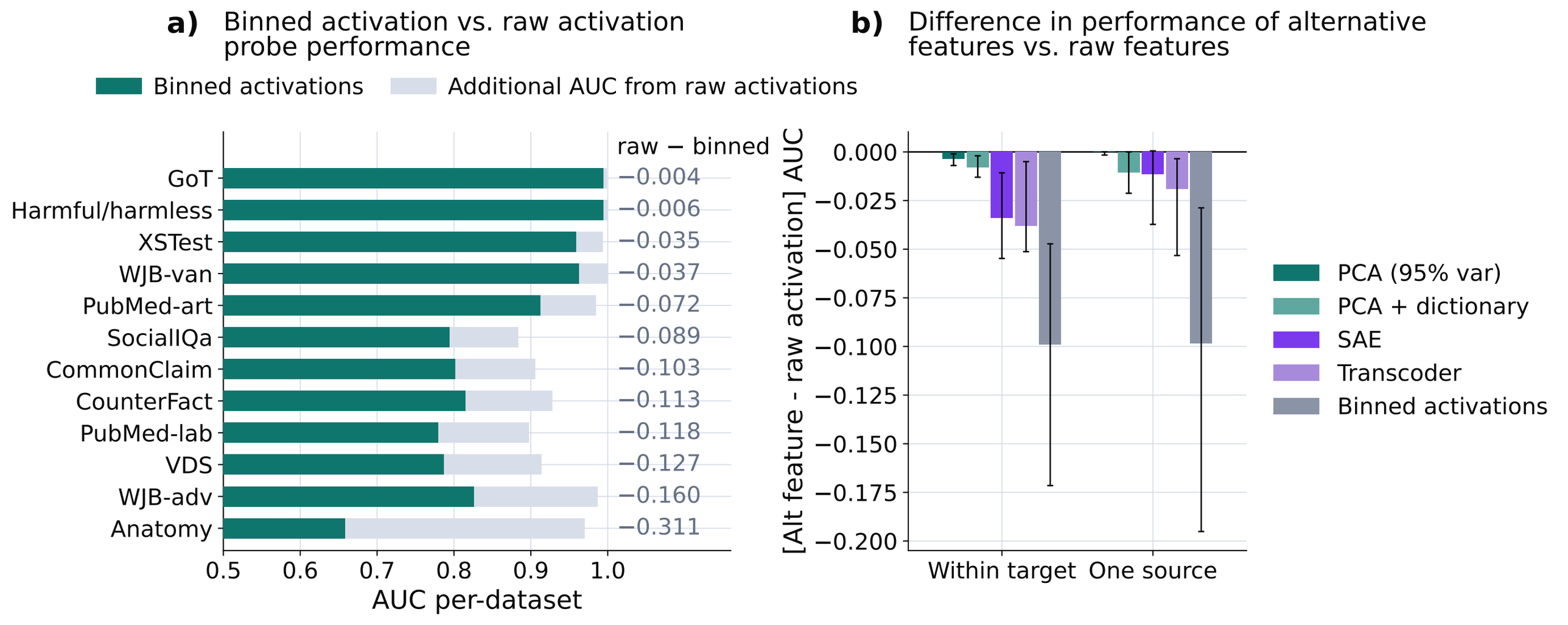
a)
Binned activation vs. raw activation probe performance
Binned activations
Additional AUC from raw activations
raw − binned
GoT −0.004
Harmful/harmless −0.006
XSTest −0.035
WJB-van −0.037
PubMed-art −0.072
SocialIQa −0.089
CommonClaim −0.103
CounterFact −0.113
PubMed-lab −0.118
VDS −0.127
WJB-adv −0.160
Anatomy −0.311
0.5
0.6
0.7
0.8
0.9
1.0
AUC per-dataset
b)
Difference in performance of alternative features vs. raw features
[Alt feature - raw activation] AUC
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
−0.150
−0.175
−0.200
Within target
One source
PCA (95% var)
PCA + dictionary
SAE
Transcoder
Binned activations